UNIVERSITY OF THE PEOPLE

CS 1101 Unit 2

Programming Assignment

Assignment Response

1. **Circumference Calculation**

The circumference of a circle is calculated by 2πr, where π = 3.14159 (rounded to five decimal places). Write a function called print_circum that takes an argument for the circle's radius and prints the circle's circumference. Call your print_circum function three times with different values for radius.

Include the following in your part 1 submission:

* The code for your print_circum function.

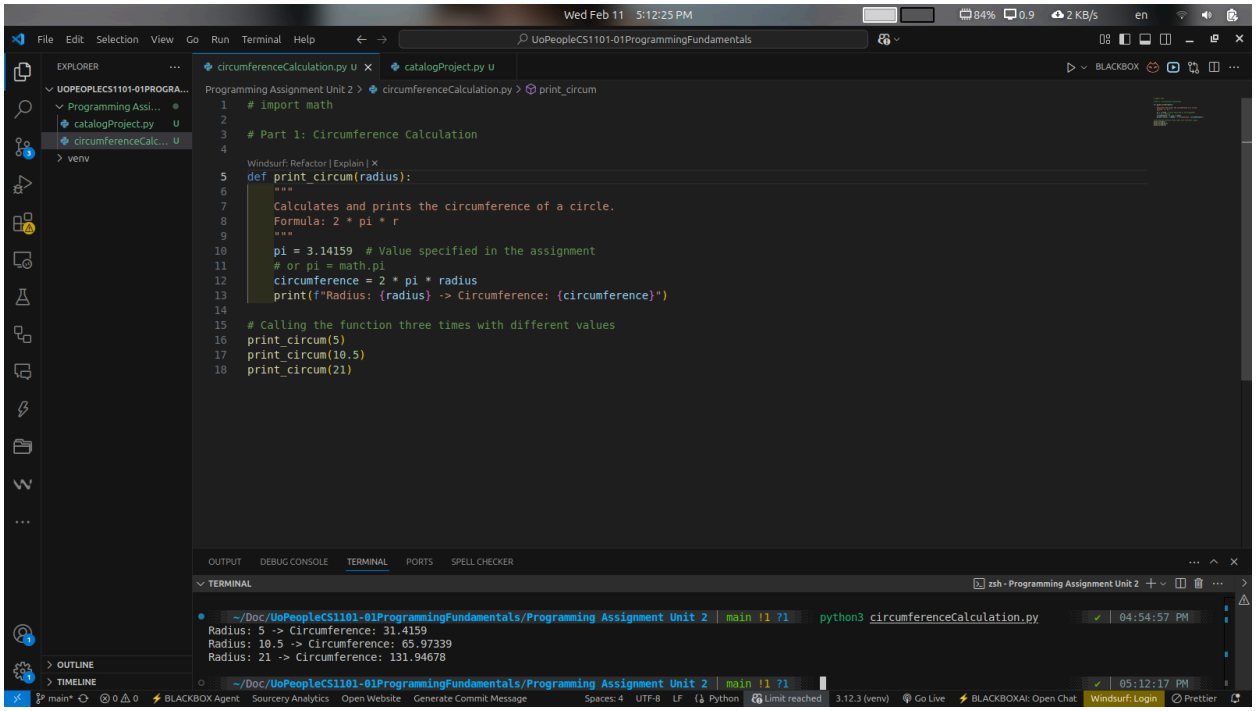
* A screenshot showing inputs and outputs to three calls of your print_circum.

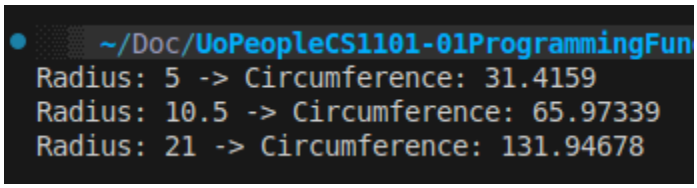
* The code and its output must be explained technically.

* The explanation can be provided before or after the code, or in the form of comments within the code.

```
# import math
def print_circum(radius):
    pi = 3.14159  # Value specified in the assignment
    # or pi = math.pi
    circumference = 2 * pi * radius
    print(f"Radius: {radius} -> Circumference: {circumference}")
```

```
# Calling the function three times with different values

print_circum(5)

print_circum(10.5)

print_circum(21)
```





This program demonstrates the use of functions to perform specific tasks. As described by Downey (2015), *"A function definition specifies the name of a new function and the sequence of statements that run when the function is called"*. The first line of my code, def

print_circum(radius):, is called the header, and the indented lines that follow are called the body. In this function, radius is a parameter. According to Downey (2015), *"Inside the function, the arguments are assigned to variables called parameters"*. When I call the function using print_circum(5), the value 5 is the argument passed to the function. I have also included notes starting with the # symbol. These are called comments, and they explain in natural language what the program is doing. For instance, I included a comment # or pi = math.pi to show that while I am using the hardcoded value 3.14159 as requested, I am aware that the math module contains a variable math.pi that stores a more precise value of pi. Because this function prints the result directly rather than returning it to the caller, it is considered a "void function".

2. Catalog Project.

Develop a catalog for a company. Assume that this company sells three different Items. The seller can sell individual items or a combination of any two items. A gift pack is a special combination that contains all three items.

Here are some special considerations:

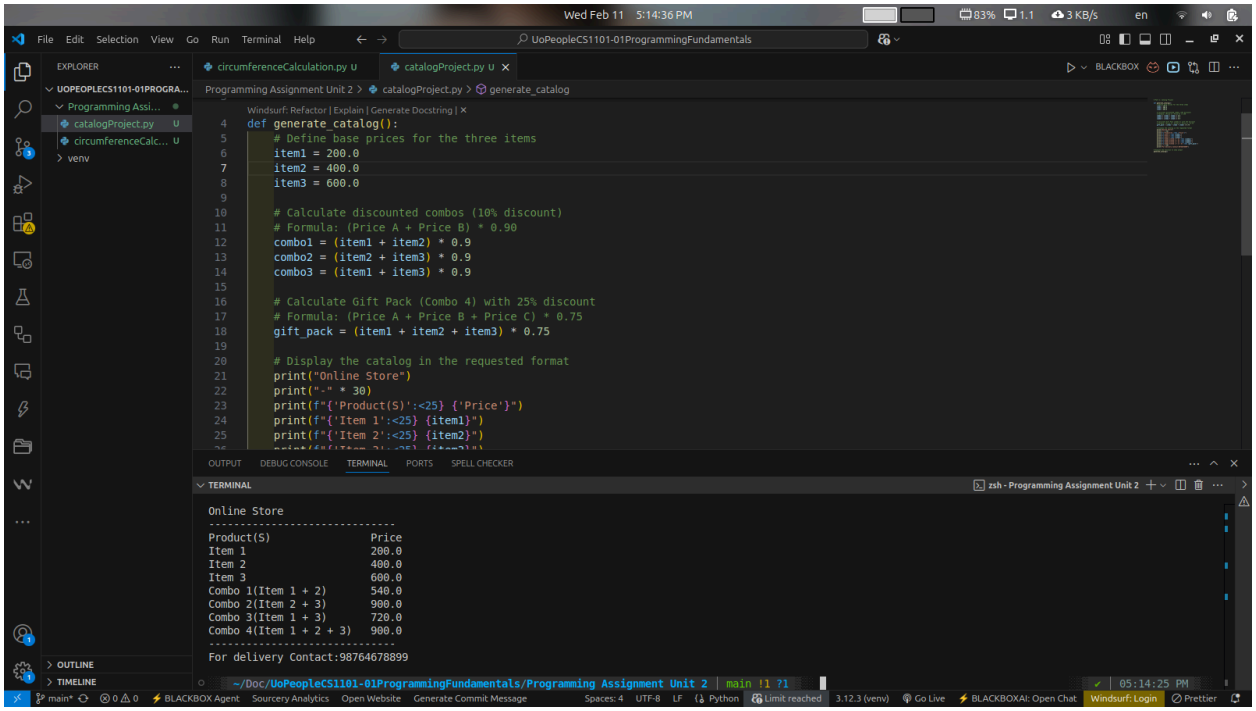
A. If a customer purchases individual items, he does not receive any discount.

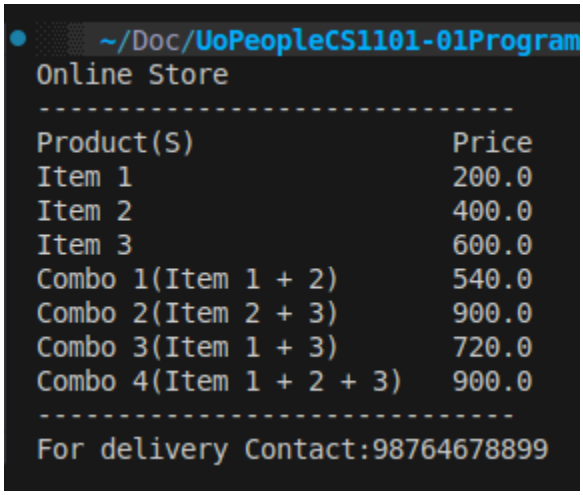
B. If a customer purchases a combo pack with two unique items, he gets a 10% discount.

C. If the customer purchases a gift pack, he gets a 25% discount.

```
def generate_catalog():
    # Define base prices for the three items
    item1 = 200.0
    item2 = 400.0
```

```
    item3 = 600.0

    # Calculate discounted combos (10% discount)
    # Formula: (Price A + Price B) * 0.90
    combo1 = (item1 + item2) * 0.9
    combo2 = (item2 + item3) * 0.9
    combo3 = (item1 + item3) * 0.9

    # Calculate Gift Pack (Combo 4) with 25% discount
    # Formula: (Price A + Price B + Price C) * 0.75
    gift_pack = (item1 + item2 + item3) * 0.75

    # Display the catalog in the requested format
    print("Online Store")
    print("-" * 30)
    print(f"{'Product(S)':<25} {'Price'}")
    print(f"{'Item 1':<25} {item1}")
    print(f"{'Item 2':<25} {item2}")
    print(f"{'Item 3':<25} {item3}")
    print(f"{'Combo 1(Item 1 + 2)':<25} {combo1}")
    print(f"{'Combo 2(Item 2 + 3)':<25} {combo2}")
    print(f"{'Combo 3(Item 1 + 3)':<25} {combo3}")
    print(f"{'Combo 4(Item 1 + 2 + 3)':<25} {gift_pack}")
    print("-" * 30)
    print("For delivery Contact:98764678899")

# Execute the function to show output
generate_catalog()
```





The generate_catalog function illustrates several key programming concepts. First, I created variables such as item1, item2, and gift_pack inside the function. As Downey (2015) notes, *"When you create a variable inside a function, it is local, which means it only exists inside that function"*. These variables store the floating-point values for the prices. To calculate the discounts, I wrote expressions using operators. *"An expression is a combination of values,*

variables, and operators" (Downey, 2015). For the combination packs, I had to be careful with the order of operations. *"Python follows mathematical conventions, often remembered by the acronym PEMDAS"* (Downey, 2015). For example, in the line (item1 + item2) * 0.9, *"Parentheses are used because they have the highest precedence"* (Downey, 2015). This forces the addition of the two items to happen before the multiplication by 0.9; otherwise, only item2 would have been multiplied. Finally, the catalog is printed using string formatting. I also used the multiplication operator on strings, such as '-' * 30, *"which performs repetition"* (Downey, 2015) to create the dividing lines.

References

Downey, A. (2015). Think Python: How to think like a computer scientist. Green Tree Press. https://greenteapress.com/thinkpython2/thinkpython2.pdf